Spheres of Dharma

A second endless golden cycle of the icosahedral solids, through the icosidodecahedron, the fullerene cage and the rhombic triacontahedron

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Version 1 · 12 October 2026 · Licence CC BY 4.0 · Data, scripts, renders and checks are included with this record and at https://github.com/DICTOR-Master/kaleidohedra.
The icosidodecahedron as the twin form, the dual triangular structures, the triangles of rhombi, the fullerene transition and the last missing layer are DICTO's ideas. Computation and checking with Claude (Anthropic).

Abstract. We describe an exact, endlessly repeating nesting of six icosahedral solids (the Ritsudō cycle), each layer filled by a few repeated pieces. With the dodecahedron at edge 1: an icosahedron of edge 1/φ, an icosidodecahedron of edge φ/2 (around it 12 pentagonal pyramids and 20 twisted triangular frusta), the dodecahedron (20 triangular pyramids), a truncated icosahedron of edge 1/φ, the fullerene cage (12 twisted pentagonal frusta and 30 Mosseri–Sadoc tetrahedra t1 at scale 1/φ), a rhombic triacontahedron of edge ½√(45 − 18√5) = 1.089814 (12 pentagonal and 20 hexagonal caps and 60 tetrahedral wedges), and an icosahedron of edge φ² (30 edge and 20 face pieces). The outer icosahedron is φ³ times the inner one, so the cycle repeats without end in both directions; one cycle has 237 pieces. All pieces of a cycle are checked to be free of overlap, with total volume exact. The construction rests on three exact relations: the icosahedron and the dodecahedron share the icosidodecahedron as their core, differing only in 12 pentagonal or 20 triangular corner pyramids; the fullerene cage wraps the dodecahedron at exactly one golden step, the dodecahedron's corners on the cage's hexagon centres; and the rhombic triacontahedron is the cage, with its 60 corners on the triacontahedron's faces, plus caps and wedges. The cycle shares its dodecahedron and its two icosahedra with the Lotus seed of the Chain of Karma, and each half of either structure fills its side of that dodecahedron exactly, so the two can be exchanged half by half at every scale. The cycle grows concentrically, one icosahedron per cycle (Nenrin growth), while the Lotus seed branches, 13 icosahedra per φ² step; the Lotus seed's 12 corner icosahedra straddle the shared dodecahedron, so they stay whole only when both halves are the Lotus seed's. The five cubes of the dodecahedron lie in the network's seams, and the five octahedra on the icosidodecahedron hold one icosahedron of edge 1 in Pacioli's golden position. Its outer shell, the rhombic triacontahedron, is the only other form found to branch like the Lotus seed (the Bodhi seed, described separately). The shells are the family of the Tsai-type cluster of icosahedral quasicrystals, here solid and exactly filled.

Checks. All 237 pieces of one cycle are pairwise free of overlap (separating axes over all face normals and edge-pair cross products), all lie inside the outer icosahedron, and their total volume equals I(φ²) to 1e-12 (scripts/sod_overlap.py). The triacontahedron's edge is exactly ½√(45 − 18√5) = 1.0898138.

Terms. The six-shell cycle is called a *Ritsudō* (律動, rhythm); its concentric mode of growth, one ring per cycle, *Nenrin* (年輪, tree rings), in contrast to the branching of the Lotus seed. Names by DICTO.

1. The shared core

Cut the corners of an icosahedron (edge 1) back to its edge midpoints and an icosidodecahedron of edge 1/2 is left, with 12 pentagonal pyramids (all edges 1/2) cut away: 1.729441 + 0.452254 = 2.181695. Do the same to a dodecahedron (edge 1): an

icosidodecahedron of edge φ/2 and 20 triangular pyramids (base φ/2, side edges 1/2): 7.326029 + 0.337090 = 7.663119. So an icosahedron of edge φ and a dodecahedron of edge 1 share one icosidodecahedron, and in their dual compound their 30 edge pairs cross at its corners.

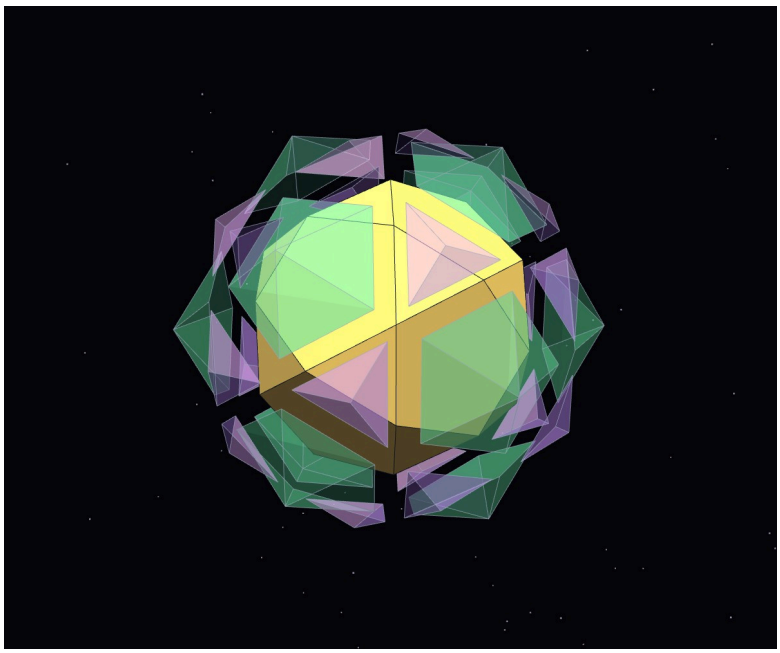


Figure 1. The shared core pulled apart: one icosidodecahedron (gold), its 12 pentagonal pyramids (green, making the icosahedron of edge φ) and 20 triangular pyramids (purple, making the dodecahedron of edge 1).

2. The dodecahedron and the fullerene cage

A truncated icosahedron of edge 1/φ contains the dodecahedron of edge 1 with the dodecahedron's 20 corners on its 20 hexagon centres. The gap (5.388544) is 12 twisted pentagonal frusta (cage pentagon, edge 1/φ, over the dodecahedron face, edge 1, turned 36°; volume 0.399864) and 30 tetrahedra with five edges 1/φ and one edge 1 (the Mosseri–Sadoc tile t1 at scale 1/φ; 0.019672).

3. The fullerene cage and the rhombic triacontahedron

With its 60 corners on the faces of a rhombic triacontahedron (cage edge 0.567101 for triacontahedron edge 1), the cage plus 12 pentagonal caps, 20 hexagonal caps (each made of the three rhombi at a three-fold corner) and 60 tetrahedral wedges along its pentagon–hexagon edges fill the triacontahedron exactly: 10.083445 + 1.704813 + 60 × 0.008708 = 12.310734.

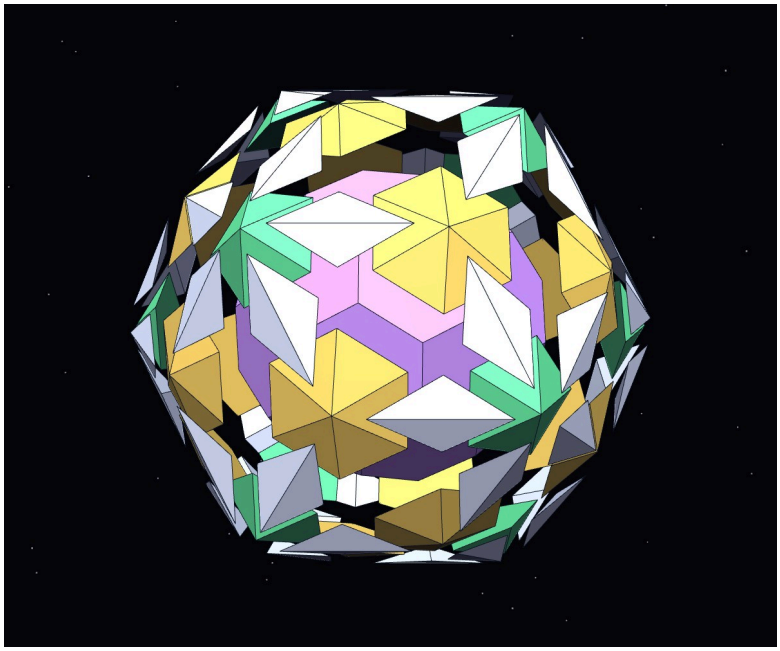


Figure 2. The fullerene cage inside the rhombic triacontahedron, pulled apart: 20 hexagonal caps (gold islands, each three rhombi), 12 pentagonal caps and 60 wedges.

4. Closing the cycle

Between the triacontahedron (edge 1.089814) and the icosahedron of edge φ^2: 30 edge pieces (each rhombus joined to the icosahedron edge it faces, 0.527864) and 20 face pieces (each icosahedron face joined to the three-fold corner below it, 0.368921). Inside the icosidodecahedron: the icosahedron of edge $1/\varphi$ with 12 pentagonal pyramids (0.197336) and 20 twisted triangular frusta (icosidodecahedron triangle over icosahedron face, side edges $\sqrt{3}/2$; 0.222148).

Shell	Edge	Filled from the shell inside by
icosahedron	$1/\varphi$	(the centre)
icosidodecahedron	$\varphi/2$	12 pentagonal pyramids + 20 twisted triangular frusta
dodecahedron	1	20 triangular pyramids
truncated icosahedron	$1/\varphi$	12 twisted pentagonal frusta + 30 t1
rhombic triacontahedron	1.089814	12 + 20 caps + 60 wedges
icosahedron	φ^2	30 edge pieces + 20 face pieces

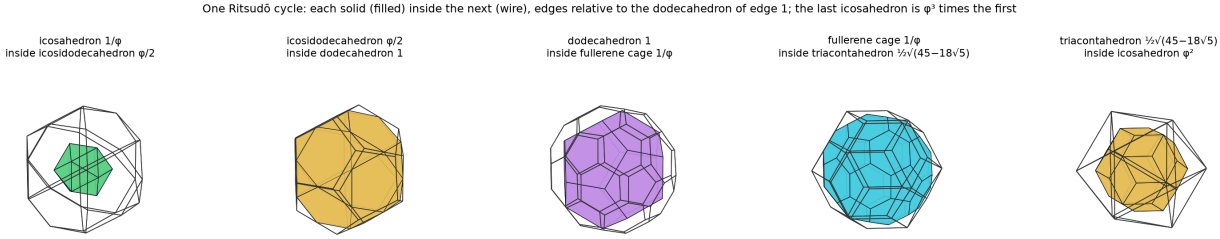


Figure 3. One Ritsudō cycle at true relative size: each solid (filled) inside the next (wire frame). The last icosahedron is φ^3 times the first.

5. The hinge with the Lotus seed

The Lotus seed (Chain of Karma, #22) fills the same icosahedron of edge φ^2 around the same dodecahedron of edge 1 and icosahedron of edge $1/\varphi$: inside the dodecahedron with 12 pentagonal pyramids, 30 AXE and 20 CLEO; outside it with 12 J11, 30 AXE and 20 FUJI. Each half of either structure fills its side exactly (volumes checked), so the four combinations are all exact and, in the Chain of Karma, every icosahedron at every scale can be filled either way. The four combinations are exact fillings, but the Lotus seed's 12 corner icosahedra straddle the dodecahedron (a J2 cap inside, a J11 outside), so they are whole only when both halves are the Lotus seed's: only the pure Lotus seed branches (13 icosahedra per φ^2 step); the Spheres of Dharma grow concentrically (one centre icosahedron per φ^3 step).

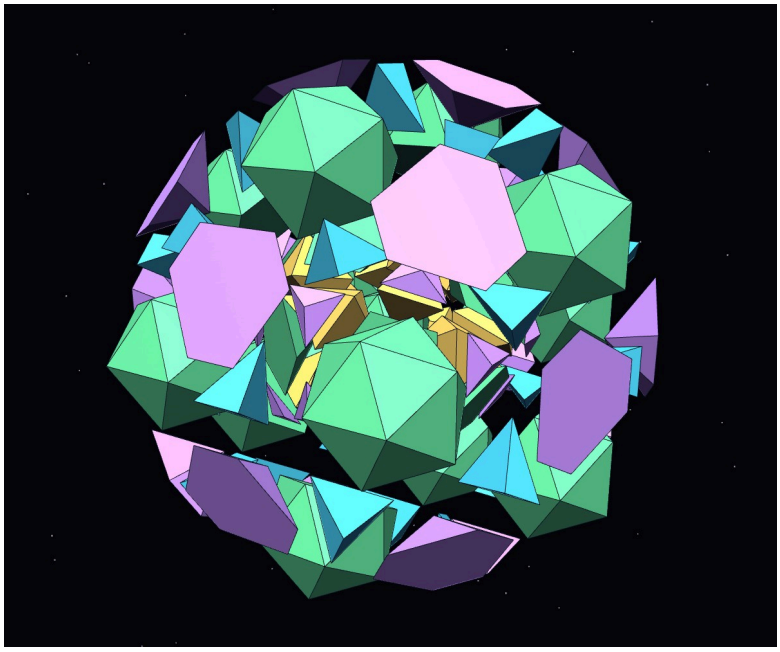


Figure 4. The hinge, pulled apart: the Spheres of Dharma inside the dodecahedron of edge 1 and the Lotus seed outside it.

6. The cube family

The five cubes inscribed in the dodecahedron have all 60 edges in the seams between pieces. The five octahedra on the icosidodecahedron's 30 corners (the compound of five octahedra) and the ten tetrahedra of the dodecahedron cut through pieces; but the five octahedra hold one icosahedron of edge 1, in the network's own turn, each of its 12 corners on an edge of every octahedron (Pacioli's golden position), suggesting an octahedral variant: icosahedron $1/\varphi \subset$ icosahedron $1 \subset$ five octahedra $\subset$ icosidodecahedron.

7. Relation to known work, and status

The dual compound of the icosahedron and dodecahedron, the compound of five octahedra and Pacioli's icosahedron in the octahedron are classical. Concentric dodecahedron, icosahedron, icosidodecahedron and triacontahedron shells form the Tsai-type cluster of icosahedral quasicrystals, as atom shells (Takakura et al. 2007); nested icosahedral point shells model viruses (Keef and Twarock 2009). We found no description of this solid, exactly filled, endlessly repeating cycle; the search is the one recorded for the Chain of Karma and is not exhaustive.

References

1. H. Takakura, C. P. Gómez, A. Yamamoto, M. de Boissieu, A. P. Tsai, Atomic structure of the binary icosahedral Yb–Cd quasicrystal, *Nature Materials* 6 (2007) 58–63.
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3. R. Mosseri and J.-F. Sadoc, in: P. H. Gaskell, J. M. Parker, E. A. Davis (eds.), The Structure of Non-Crystalline Materials 1982 (Taylor & Francis, London, 1983) p. 137; P. Kramer, Non-periodic central space filling with icosahedral symmetry using copies of seven elementary cells, *Acta Cryst.* A38 (1982) 257–264 (the tile t1).
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